

Bacterial Classification

Microbial Taxonomy

Classification Systems

Levels of Classification

Definition of “Species”

Nomenclature

Useful Properties in Microbial Classification

Microbial Phylogeny

Taxonomy: the Science of Classification

The science of classifying organisms

Provides universal names for organisms

Provides a reference for identifying organisms

Groupings of organisms

WHY Classify?

Establish criteria for ID

Arrange related organisms into groups

Provide information about evolution of organisms

Systematics or phylogeny: The study of the evolutionary history of organisms.

To identify all species of life on Earth

Linnaeus – (1707-1778) Father of modern taxonomy

Created Binomial nomenclature

2 names- *Genus-species*

Classification Systems

Taxonomy:

Classification of living organisms into groups

Phylogenetic Classification System:

Groups reflect genetic similarity and evolutionary relatedness

Phenetic Classification System:

Groups do not necessarily reflect genetic similarity or evolutionary relatedness. Instead, groups are based on convenient, observable characteristics.

Levels of Classification

Taxon:

A group or “level” of classification

Hierarchical; broad divisions are divided up into smaller divisions:

- Kingdom (Not used by most bacteriologists)

- Phylum (Called “Division” by botanists)

- Class

- Order

- Family

- Genus (plural: Genera)

- Species (Both singular & plural)

- Subspecies/Strain

Definition of “Species”

The “basic unit” of taxonomy, representing a specific, recognized type of organism

For sexually reproducing organisms, a fundamental definition of “species” has been reproductive compatibility

This definition fails for many microbial species (including bacteria), because they do not reproduce sexually

Definition of “Species”

Classic definition: A collection of microbial strains that share many properties and differ significantly from other groups of strains

Species are identified by comparison with known “type strains”: well-characterized pure cultures; references for the identification of unknowns

Definition of “Species”

Definition of “species” in microbiology (cont.):

Strain:

A population of microbes descended from a single individual or pure culture

Different strains represent genetic variability within a species

Biovars: Strains that differ in biochemical or physiological differences

Morphovars: Strains that vary in morphology

Serovars: Strains that vary in their antigenic properties

Nomenclature

Scientific name (Systematic Name)

Binomial System of Nomenclature

Genus name + species name

Italicized or underlined

Genus name is capitalized and may be abbreviated

Species name is never abbreviated

A genus name may be used alone to indicate a genus group; a species name is never used alone

eg: *Bacillus subtilis*

B. subtilis

Nomenclature

Common or descriptive names
(trivial names)

Names for organisms that may be in common usage, but are not taxonomic names

eg: tubercle bacillus

(Mycobacterium tuberculosis)

meningococcus

(Neisseria meningitidis)

Group A streptococcus

(Streptococcus pyogenes)

Useful Properties in Classification

Colony morphology

Cell shape & arrangement

Cell wall structure (Gram staining)

Special cellular structures

Biochemical characteristics

Serological Tests

- Use group specific antiserum isolated from the plasma of animals that have been sensitized to the organism

The antiserum contains antibody proteins that react with antigens on the unknown organism.

The reaction can be detected by examining agglutination or by using sera labeled with colorimetric or fluorescent labels

Serological Tests

Advantages:

Highly specific

Does not usually require the organism to be isolated into pure culture

Can be used to identify organisms that can't be grown on medium

Higher G + C gives a higher melting temperature

Nucleic acid hybridization

- By mixing ssDNA from two different species and determining the percentage of the DNA that can form dsDNA hybrids
- The greater the percent hybridization, the closer the species

Microbial Phylogeny

Bergey's Manual of Systematic Bacteriology

In 1927, David Bergey & colleagues published *Bergey's Manual of Determinative Bacteriology*, a manual that grouped bacteria into phenetic groups, used in identification of unknowns.

In 1984, a more detailed work entitled *Bergey's Manual of Systematic Bacteriology* was published, still primarily phenetic in its classification.

Microbial Phylogeny

Bergey's Manual of Systematic Bacteriology

- Publication of the second edition of *Bergey's Manual of Systematic Bacteriology* was begun in 2001.
- The 2nd edition gives the most up-to-date phylogenetic classification of prokaryotic organisms, including both eubacteria and archaea.
- When it is completed, it will consist of 5 volumes.
- The classification in *Bergey's Manual* is accepted by most microbiologists as the best consensus for prokaryotic taxonomy.

5 Kingdom System or 6 Kingdom System

- Kingdom Monera - Eubacteria - new bacteria -
Prokarya - Archaeobacteria – old bacteria
- Kingdom Protista- 1 celled organisms- Eukarya
- Kingdom Fungi- multicellular fungi/yeast- Eukarya
- Kingdom Plantae- photosynthetic plants- Eukarya
- Kingdom Animalia- animals from zygote- Eukarya

The Three-Domain System

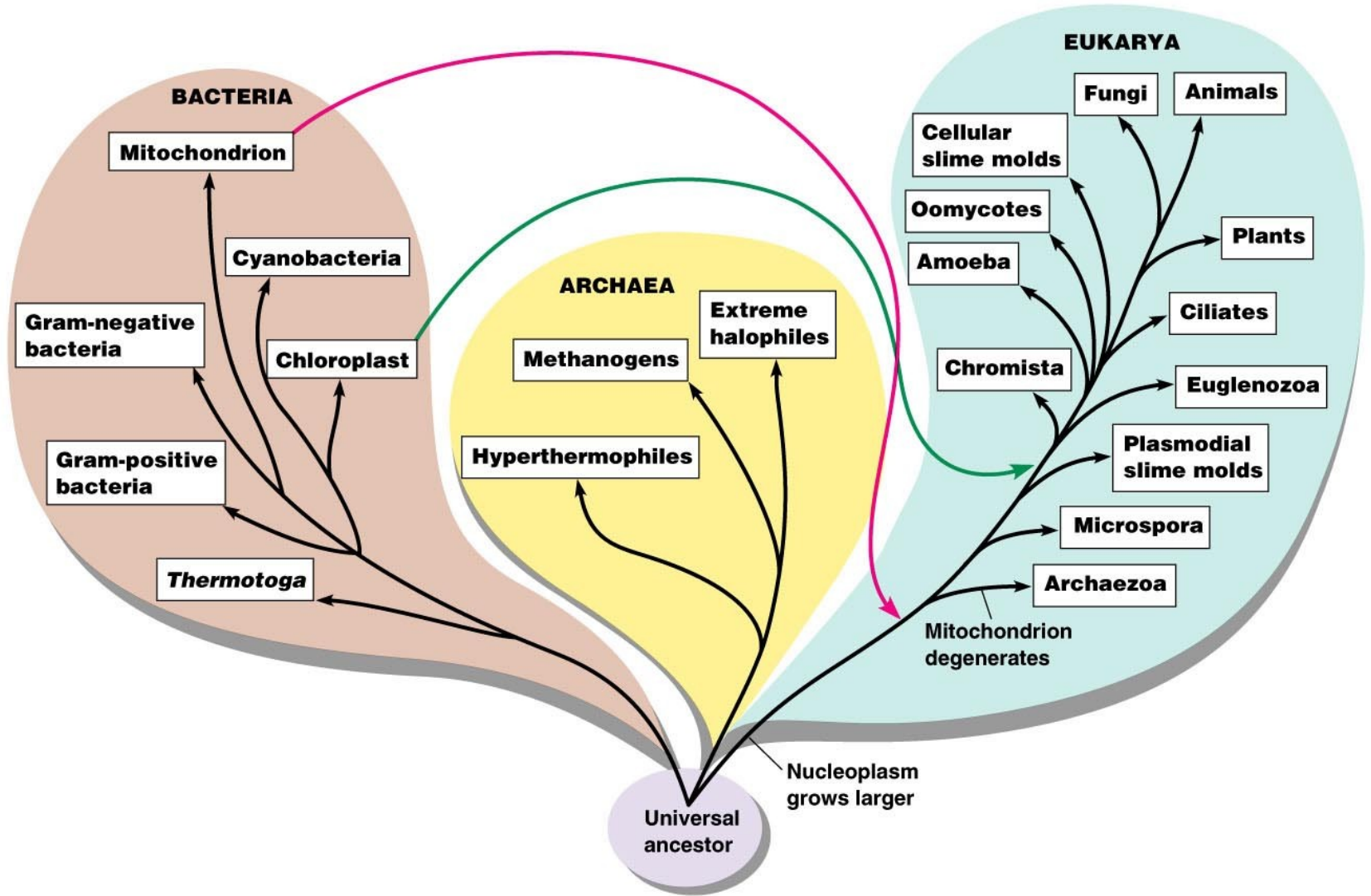
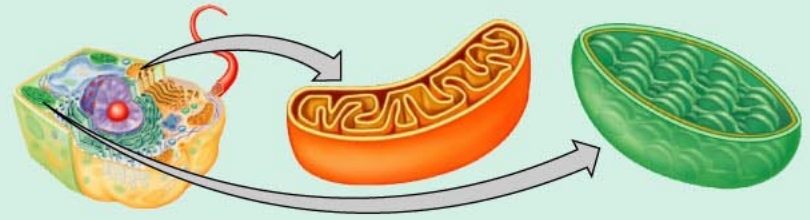
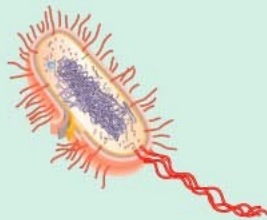


Figure 10.1

TABLE 10.2

Prokaryotic Cells and Eukaryotic Organelles Compared

	Prokaryotic Cell	Eukaryotic Cell	Eukaryotic Organelles (Mitochondria and Chloroplasts)
DNA	One circular; some two circular; some linear	Linear	Circular
Histones	In archaea	Yes	No
First Amino Acid in Protein Synthesis	Formylmethionine	Methionine	Formylmethionine
Ribosomes	70S	80S	70S
Growth	Binary fission	Mitosis	Binary fission



A Comparison of the More Notable Classification Systems of Living Organisms

Haeckel (1894)	Whittaker (1959)	Woese (1977)	Woese (1990)		
Three kingdoms	Five kingdoms	Six kingdoms	Three domains		
Protista	Monera	Eubacteria	Bacteria		
Plantae	Protista	Archaeobacteria	Archaea		
Animalia	Fungi	Protista	Eukarya	Plantae	Fungi
	Animalia	Plantae			
		Animalia			

Similarities:

Archaea have generally the same shape, size, and appearance as bacteria.

Like bacteria, Archaea multiply by binary fission and move primarily by means of flagella.

These morphological similarities can make it difficult to visually tell a bacterium and an archaean apart.

Additionally, archaea are more phylogenetically similar to eukarya than bacteria is to either of them.

Differences

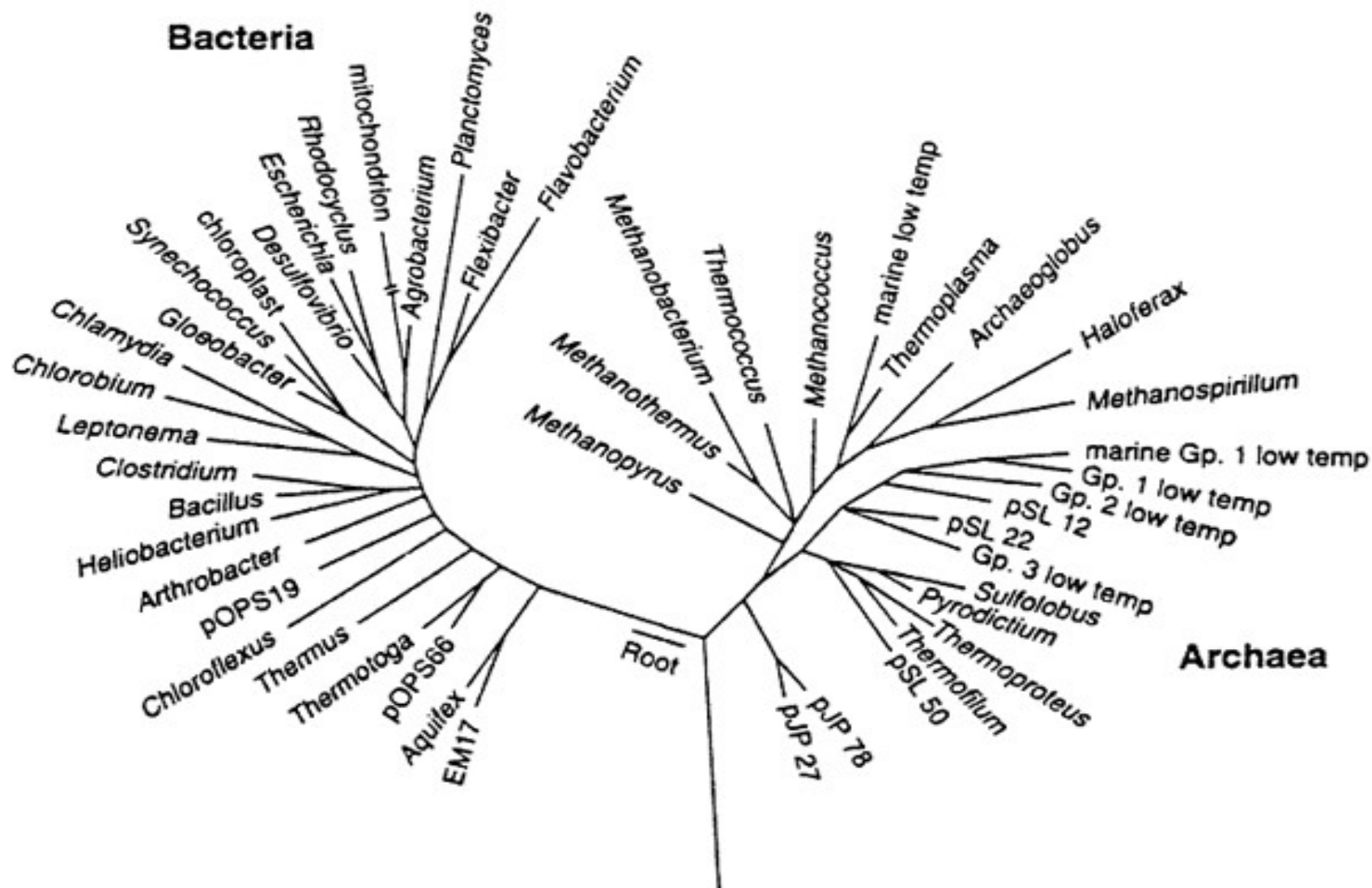
Unlike bacteria, archaea cell walls do not contain peptidoglycan.

Chemistry of cell walls is different from bacteria: archaeal cell walls don't contain peptidoglycan. Archaea have various types of cell wall including pseudopeptidoglycan

No D-amino acids

Archaea have different membrane lipid bonding compared to Bacteria and Eucarya. Archaea membrane lipids have ether bonds instead of ester bonds.

Archaeal lipids lack the fatty acids found in Bacteria and Eukaryotes and instead have side chains composed of repeating units of isoprene.



Explanation of the diagram

- The Phylogenetic Tree of Life based on Comparative ssrRNA Sequencing.
- The Tree shows the procaryotes in two Domains, Archaea and Bacteria.
- At a taxonomic level, most organisms at the tips of the Archaeal branches represent a unique Order; most organisms at the tips of the bacterial branches are classified into a unique Phylum.
- On the Archaeal limb, the three physiological groups are evident in the names: "thermo" and "pyro" for the extreme thermophiles; "methano" for the methanogens; and "halophiles" for the extreme halophiles.
- The most important, best known, and diverse groups (phyla) branching off of the Bacterial limb are the Cyanobacteria, Proteobacteria and Gram positives.

Microbial Phylogeny

Domains

Based on the research of Woese and others in the 1980s and 1990s, most biologists divide all living organisms into 3 domains:

Domain *Archaea*

Domain *Bacteria*

Domain *Eucarya*

rRNA sequence data suggests that *Archaea* & *Eucarya* may share a more recent common ancestor with each other than with *Bacteria*

The Three-Domain System

TABLE 10.1

Some Characteristics of Archaea, Bacteria, and Eukarya

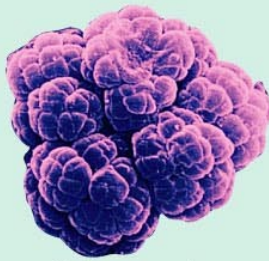
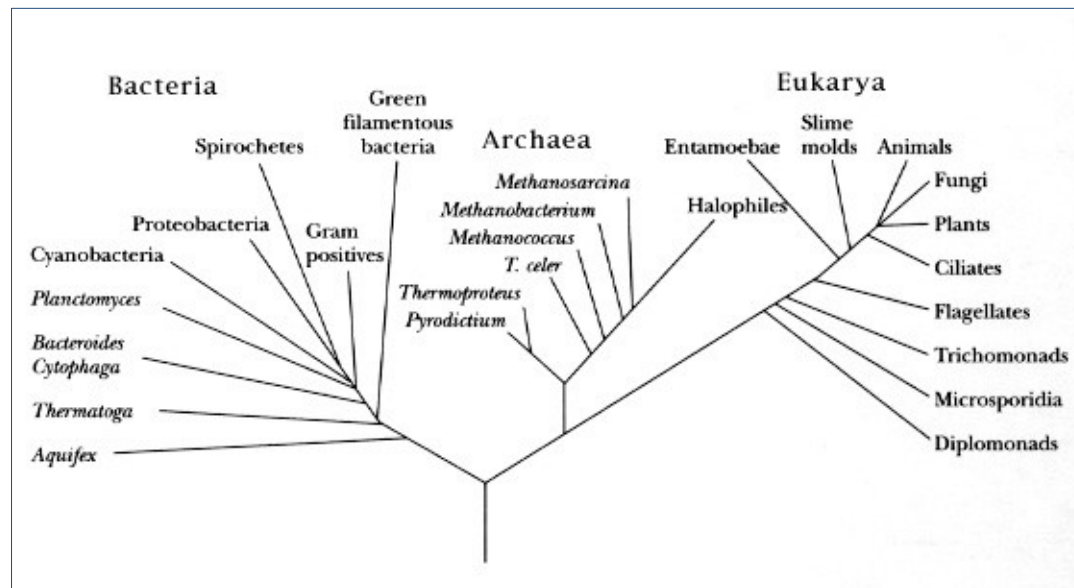
	Archaea	Bacteria	Eukarya
	 <i>Methanosarcina</i> SEM 10 μm	 <i>E. coli</i> SEM 1 μm	 <i>Amoeba</i> SEM 10 μm
Cell Type	Prokaryotic	Prokaryotic	Eukaryotic
Cell Wall	Varies in composition; contains no peptidoglycan	Contains peptidoglycan	Varies in composition; contains carbohydrates
Membrane Lipids	Composed of branched carbon chains attached to glycerol by ether linkage	Composed of straight carbon chains attached to glycerol by ester linkage	Composed of straight carbon chains attached to glycerol by ester linkage
First Amino Acid in Protein Synthesis	Methionine	Formylmethionine	Methionine
Antibiotic Sensitivity	No	Yes	No
rRNA Loop*	Lacking	Present	Lacking
Common Arm of tRNA[†]	Lacking	Present	Present
*Binds to ribosomal protein; found in all bacteria. [†]A sequence of bases in tRNA found in all eukaryotes and bacteria: guanine-thymine-pseudouridine-cytosine-guanine.			

Table 10.1

The Phylogenetic Tree of Life based on Comparative ssrRNA* Sequencing



*the nucleotide sequences of the small subunit ribosomal RNA.

Microbial Phylogeny

- Domains (cont.)
 - Many microbiologists reject the “kingdom” designation.
 - Each domain is divided into phyla, phyla into classes. etc.
 - There is often great metabolic and ecological diversity among the members of a group, perhaps reflecting parallel evolution of such things as fermentation pathways, photosynthetic pathways, etc.

Microbial Phylogeny

- Phylogeny of domain *Archaea*
 - Based primarily on rRNA sequence data, domain *Archaea* is divided into two phyla:
 - Phylum *Crenarchaeota*
 - Originally containing thermophilic and hyperthermophilic sulfur-metabolizing archaea
 - Recently discovered *Crenarchaeota* are inhibited by sulfur & grow at lower temperatures
 - Phylum *Euryarchaeota*
 - Contains primarily methanogenic archaea, halophilic archaea, and thermophilic, sulfur-reducing archaea

Microbial Phylogeny

Phylogeny of domain *Bacteria*

- The 2nd edition of *Bergey's Manual of Systematic Bacteriology* divides domain *Bacteria* into 23 phyla. Nine of the more notable phyla are described here.
 - Phylum *Aquiflexa*
 - The earliest “deepest” branch of the *Bacteria*
 - Contains genera *Aquiflex* and *Hydrogenobacter* that can obtain energy from hydrogen via chemolithotrophic pathways

Microbial Phylogeny

Phylogeny of domain *Bacteria* (cont.)

- Phylum *Cyanobacteria*
 - Oxygenic photosynthetic bacteria
- Phylum *Chlorobi*
 - The “green sulfur bacteria”
 - Anoxygenic photosynthesis
 - Includes genus *Chlorobium*

Microbial Phylogeny

- Phylogeny of domain *Bacteria* (cont.)
 - Phylum *Proteobacteria*
 - The largest group of gram-negative bacteria
 - Extremely complex group, with over 400 genera and 1300 named species
 - All major nutritional types are represented: phototrophy, heterotrophy, and several types of chemolithotrophy
 - Sometimes called the “purple bacteria,” although very few are purple; the term refers to a hypothetical purple photosynthetic bacterium from which the group is believed to have evolved

Microbial Phylogeny

- Phylogeny of domain *Bacteria* (cont.)
 - Phylum *Proteobacteria* (cont.)
 - Divided into 5 classes: *Alphaproteobacteria*, *Betaproteobacteria*, *Gammaproteobacteria*, *Deltaproteobacteria*, *Epsilonproteobacteria*

Microbial Phylogeny

Phylogeny of domain *Bacteria* (cont.)

- Phylum *Proteobacteria* (cont.)
 - Significant groups and genera include:
 - » Photosynthetic genera such as *Rhodospirillum* (a purple non-sulfur bacterium) and *Chromatium* (a purple sulfur bacterium)
 - » Sulfur chemolithotrophs, genera *Thiobacillus* and *Beggiatoa*
 - » Nitrogen chemolithotrophs (nitrifying bacteria), genera *Nitrobacter* and *Nitrosomonas*
 - » Other chemolithotrophs, genera *Alcaligenes*, *Methylobacillus*, *Burkholderia*

Microbial Phylogeny

Phylogeny of domain *Bacteria* (cont.)

- Phylum *Proteobacteria* (cont.)
 - Significant groups and genera include:
 - » The family *Enterobacteriaceae*, the “gram-negative enteric bacteria,” which includes genera *Escherichia*, *Proteus*, *Enterobacter*, *Klebsiella*, *Salmonella*, *Shigella*, *Serratia*, and others
 - » The family *Pseudomonadaceae*, which includes genus *Pseudomonas* and related genera
 - » Other medically important *Proteobacteria* include genera *Haemophilus*, *Vibrio*, *Camphylobacter*, *Helicobacter*, *Rickettsia*, *Brucella*

Microbial Phylogeny

- Phylogeny of domain *Bacteria* (cont.)
 - Phylum *Firmicutes*
 - “Low G + C gram-positive” bacteria
 - Divided into 3 classes
 - » Class I – Clostridia; includes genera *Clostridium* and *Desulfotomaculatum*, and others
 - » Class II – *Mollicutes*; bacteria in this class cannot make peptidoglycan and lack cell walls; includes genera *Mycoplasma*, *Ureaplasma*, and others
 - » Class III – Bacilli; includes genera *Bacillus*, *Lactobacillus*, *Streptococcus*, *Lactococcus*, *Geobacillus*, *Enterococcus*, *Listeria*, *Staphylococcus*, and others

Microbial Phylogeny

Phylogeny of domain *Bacteria* (cont.)

- Phylum *Actinobacteria*
 - “High G + C gram-positive” bacteria
 - Includes genera *Actinomyces*, *Streptomyces*, *Corynebacterium*, *Micrococcus*, *Mycobacterium*, *Propionibacterium*
- Phylum *Chlamidiae*
 - Small phylum containing the genus *Chlamydia*

Microbial Phylogeny

- Phylogeny of domain *Bacteria* (cont.)
 - Phylum *Spirochaetes*
 - The spirochaetes
 - Characterized by flexible, helical cells with a modified outer membrane (the outer sheath) and modified flagella (axial filaments) located within the outer sheath
 - Important pathogenic genera include *Treponema*, *Borrelia*, and *Leptospira*
 - Phylum *Bacteroidetes*
 - Includes genera *Bacteroides*, *Flavobacterium*, *Flexibacter*, and *Cytophyga*; *Flexibacter* and *Cytophyga* are motile by means of “gliding motility”

Microbial Phylogeny

Phylogeny of domain *Eucarya*

- The domain *Eucarya* is divided into four kingdoms by most biologists:

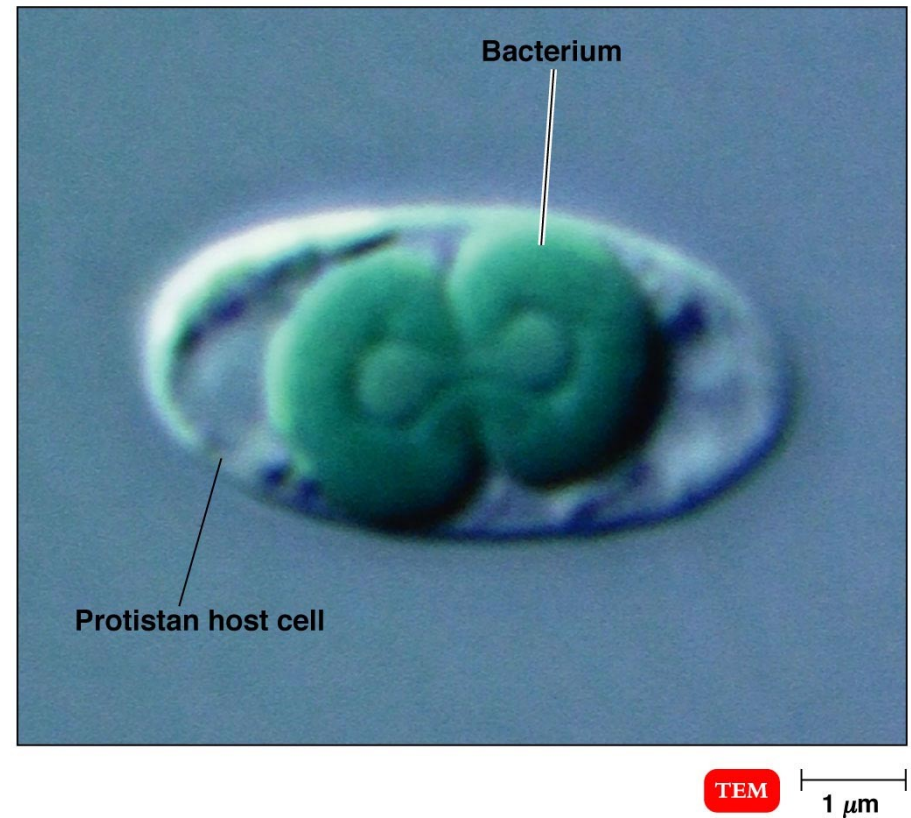
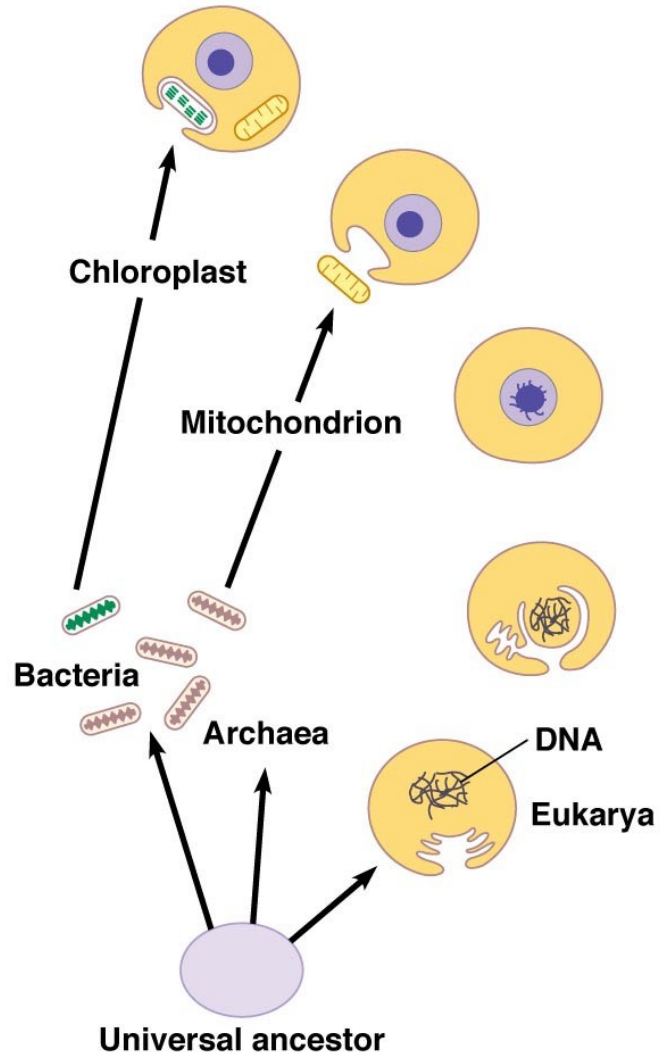
Kingdom *Protista*, including the protozoa and algae

Kingdom *Fungi*, the fungi (molds, yeast, and fleshy fungi)

Kingdom *Animalia*, the multicellular animals

Kingdom *Plantae*, the multicellular plants

Endosymbiotic Theory



Species Definition

- Eukaryotic species: A group of closely related organisms that breed among themselves
- Prokaryotic species: A population of cells with similar characteristics
 - Clone: Population of cells derived from a single cell
 - Strain: Genetically different cells within a clone
- Viral species: Population of viruses with similar characteristics that occupies a particular ecological niche

Domain Eukarya

Animalia: Multicellular; no cell walls;
chemoheterotrophic

Plantae: Multicellular; cellulose cell walls;
usually photoautotrophic

Fungi: Chemoheterotrophic; unicellular or
multicellular; cell walls of chitin; develop from
spores or hyphal fragments

Protista: A catchall for eukaryotic organisms that
do not fit other kingdoms

Classification of Viruses

Acellular infectious agents

DNA and protein coat

Classified by chemical and physical characteristics

Shape, Symmetry of protein coat

Presence of envelope, Enzymes and/or tail,

Virology- study of viruses is important because:

- 1) recognized branch of Microbiology
- 2) Concern to health scientists. Many cause diseases in humans, animals, plants and other microbes

Identification Methods

- Morphological characteristics: Useful for identifying eukaryotes
- Differential staining: Gram staining, acid-fast staining
- Biochemical tests: Determines presence of bacterial enzymes

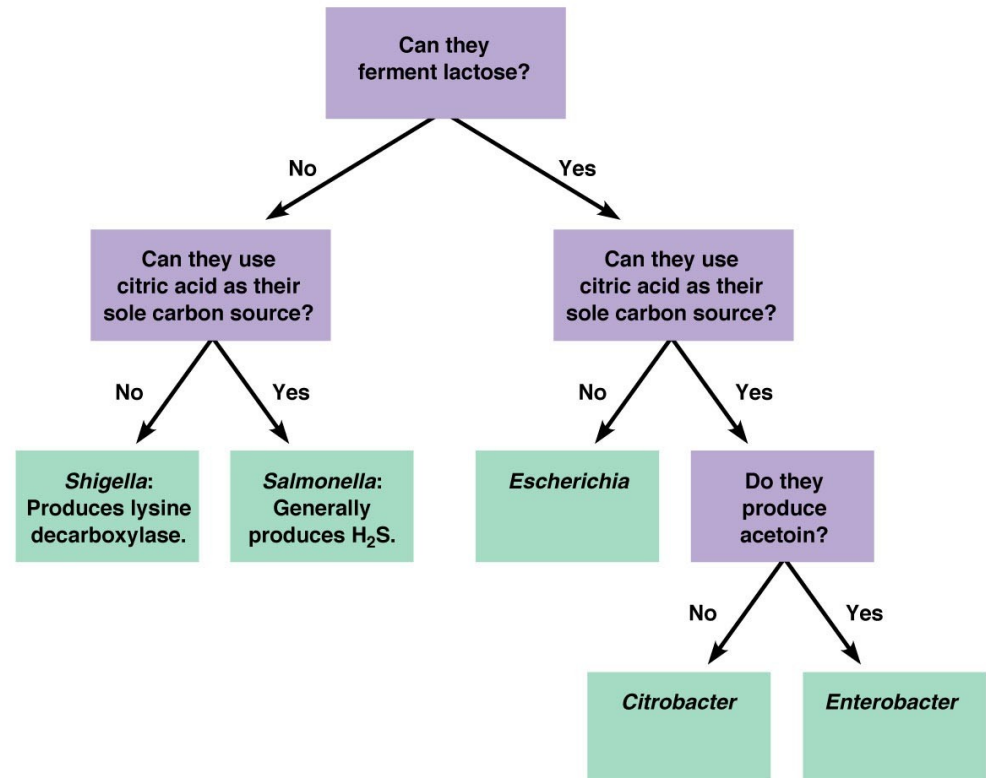


Figure 10.8

Culturing Bacteria

Microbiological culture is a principal tool used to diagnose infectious disease.

In a microbial culture, a growth medium is provided for a specific agent.

A sample taken from potentially diseased tissue or fluid is then tested for the presence of an infectious agent able to grow within that medium.

Biochemical Tests

Biochemical tests used in the identification of infectious agents include the detection of metabolic or enzymatic products characteristic of a particular infectious agent.

Since bacteria ferment carbohydrates in patterns characteristic of their genus and species.

Immunological Tests

Immunologic or Serological methods are highly sensitive, specific and often extremely rapid tests used to identify microorganisms.

These tests are based upon the ability of an antibody to bind specifically to an antigen. The antigen, usually a protein or carbohydrate made by an infectious agent, is bound by the antibody.

Molecular Diagnostics

- Technologies based upon the polymerase chain reaction (PCR) method will become nearly ubiquitous gold standards of diagnostics of the near future, for several reasons.
- First, the catalog of infectious agents has grown to the point that virtually all of the significant infectious agents of the human population have been identified.
- Second, an infectious agent must grow within the human body to cause disease; essentially it must amplify its own nucleic acids in order to cause a disease.
- This amplification of nucleic acid in infected tissue offers an opportunity to detect the infectious agent by using PCR.
- Third, the essential tools for directing PCR, primers, are derived from the genomes of infectious agents, and with time those genomes will be known, if they are not already.